

Heaven's Light Is Our Guide



Department of Mechatronics Engineering, RUET

MTE 1102

Mechatronic Systems Sessional

Project Proposal On: Automatic Irrigation System

Submitted By:

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Introduction:

The Automatic Irrigation System is a self-operated irrigation system. The system irrigates plants and farmlands according to soil conditions. It allows precise and efficient water distribution without human interactions saving precious time for farmers and gardeners. The system automatically terminates irrigation process when enough water gets supplied. Thus, this system is very effective in water scarce environments where it can strictly maintain the delivery of only the required amount of water.

Initiation of irrigation is based on soil moisture using hydrometers. When the hydrometers detect moisture deficiency of the soil, it sends signals to the microcontroller (Arduino Board). The microcontroller then opens the valves of respective positions and water starts to flow. When the hydrometers sense sufficient moisture and sends the data to the microcontroller (Arduino Board) , the valves gets closed and water flow gets terminated. The process runs continuously all day long, providing water whenever there is a need.

In this modern era of sustainability, this system provides a smart and sustainable production process. It reduces responsibility and time consumptions on farmers and lets them focus on other tasks and helps increase productivity. It prevents water waste and over/under irrigation. It reduces required manpower, and the installation process is relatively simple and requires no specific expertise. The system is relatively cheap and affordable.

Objectives:

1. By This automatic irrigation system We can automatically monitor our land moisture, use limiting water, conserve the water, save our time, and manpower.
2. We can easily save our crops from drought, Control land moisture favor to crops, which is helpful to increase our productivity.

Methodology:

The data pins of soil moisture sensor is connected to A1 pin of Arduino board and its probe is placed near the roots of the plant and Vcc pin is connected to 5 V pin of board and GND of sensor to GND of board. Probe of soil moisture sensor should be inserted near the roots of the plant. 6V pump is connected to a BC547 transistor. It is connected in this way because the amount of power that is given to the motor directly by Arduino is not sufficient to run it. The base of transistor is connected to pin 10 of board.

Working is as follows:

The probe connected to the sensor sends some amount of current into the soil. If the soil is having high moisture content, then it will allow the current to pass through it easily. Output pin will be low and motor will remain OFF. If the soil has less moisture content, then it will not allow the current to flow through. Output pin will be high, and motor will remain ON. By comparing the difference in the rate of flow of current, the moisture in the soil is calculated. The moisture sensor measures according to the code transferred into the Arduino board.

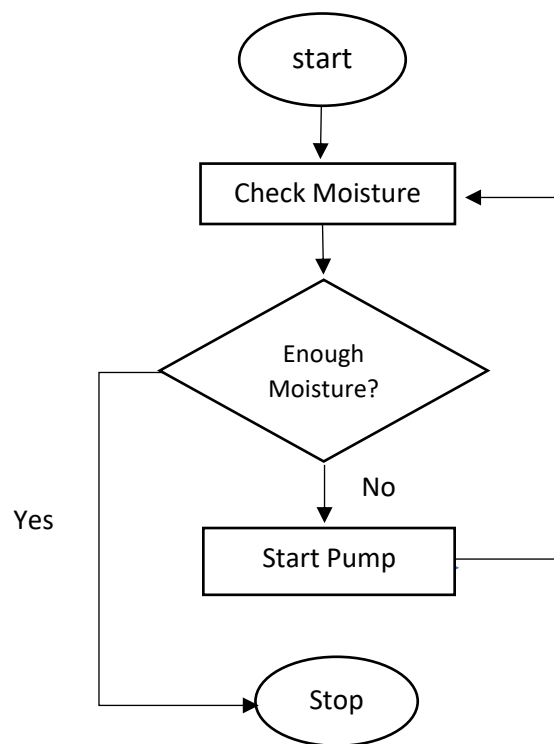


Fig: Flow chart of automatic irrigation system

Circuit Diagram:

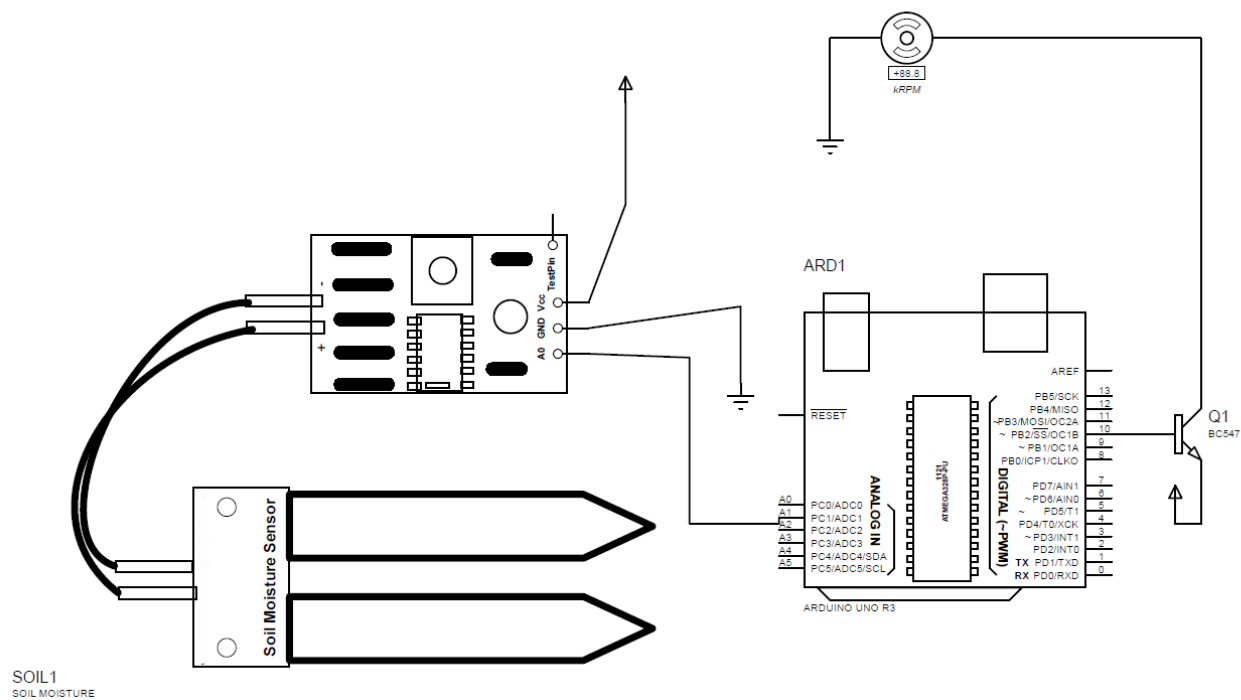


Fig: circuit diagram of an automatic water irrigation system

Required components:

- Component Name : Arduino Nano R3
Mode : R3
Cost : 860 taka



Fig: Arduino Nano R3

2. Component Name : Submersible Pump
Model :
Cost : 150 taka



Fig: 5V submersible pump

3. Component Name : Soil Moisture Sensor
Model :
Cost : 50 taka

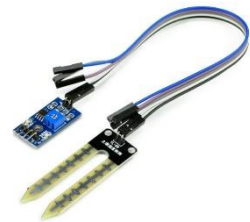


Fig: Soil Moisture Sensor

4. Component Name : 12V Valve
Model :
Cost : 360 taka



Fig:12V valve

5. Component Name : Water pipe
Model :
Cost : 50 taka



Fig: Water Pipe

6. Component Name : Jumper Wire
Model :
Cost : 50 taka



Fig: Jumper wire

7. Component Name : 12V Battery
Model :
Cost : 460 taka



Fig: 12V Battery

Expected Outcomes:

The Automatic Irrigation system brings automation into the existing irrigation practices and improves upon them. This system due to its autonomous selective water supply capability can drastically increase water conservation and can achieve timely distribution of water. It can be used to control irrigation for farmlands, fruit & vegetable gardens, indoor/outdoor potted plants etc. It is especially useful when used with plants with different water demands.

Overall, this system offers maximum water savings, reduced labor cost, easy installation and efficient management.

Discussion and Conclusion:

From the Automatic Irrigation System, irrigation becomes easy, accurate and practical with the idea above shared and can be implemented in agricultural fields in future to promote agriculture to next level. The moisture sensors measure the moisture level in different plants and the system works through the programmed function. By using this

system, it will be a clever approach to watering that means the plants will get the exact amount of water they need. Also by using this system we don't need to think when to water the plant, how often and for how long, as it is an automatic irrigation system, everything will be programmed. This system will save labor cost and also time.